SCOT: Complete Proofs and Supporting Definitions

Polong Chen, Jianfeng Du, Hai Wan, Weilin Luo, Yanan Liu, Junhua Ma, and Rongzhen Ye

***Abstract*—This self-contained technical document specifies the graph compression, one-hop rule construction, and four conjunctive-query evaluation modes of SCOT. It gives the definitions, notation, procedure descriptions, formal statements, and complete proofs needed to verify closure preservation, one-hop completeness, and exact query answering independently.**

I. Setting and Notation

A term is a variable or a constant. A ground atom contains no variables. A graph G is a finite set of ground unary or binary atoms, also called facts. All facts, relations, and query answers have set semantics. A plain Datalog (PD) rule is a function-free, universally quantified, range-restricted definite clause $\gamma : H_\gamma \leftarrow \beta_1 \wedge \cdots \wedge \beta_s$, where $s \geq 1$ and each head variable appears in its body. Constants, repeated variables, and recursion are allowed. The body set $B_\gamma = \{\beta_1, \ldots, \beta_s\}$ identifies duplicate atoms, whereas the list L_γ retains indexed occurrences during unfolding. The finite input rule set is Σ.

A substitution maps variables to terms and extends to atoms, rule bodies, and rules. A ground substitution assigns constants to all variables of the relevant rule. For a rule set Γ and graph F, define

$$\tau_{\Gamma,F} = \{\sigma(H_\gamma) : \gamma \in \Gamma,\ \sigma(B_\gamma) \subseteq F,\ \sigma \text{ ground}\}. \quad (1)$$

Starting with $F^{(0)} = F$, repeatedly set $F^{(i+1)} = F^{(i)} \cup \tau_{\Sigma,F^{(i)}}$. The finite least fixpoint is $\Delta_{F,\Sigma}$. Finiteness follows from the finite active domain and absence of function symbols. Closure is monotone and idempotent. The notation $\Sigma \models \gamma$ means that every model of Σ satisfies every ground instance of γ. A graph is rule-closed if $G = \Delta_{G,\Sigma}$.

Two atoms unify when a substitution makes them identical. A most general unifier (MGU) μ factors every unifier ρ on those atoms as $\rho = \theta \circ \mu$. We use a relevant MGU that fixes variables outside the two atoms and rename independent rule copies apart. A rule γ' subsumes γ if a substitution θ satisfies $\theta(H_{\gamma'}) = H_\gamma$ and $\theta(B_{\gamma'}) \subseteq B_\gamma$. An identity rule for predicate p is $p(\vec{Z}) \leftarrow p(\vec{Z})$ with distinct variables of the appropriate arity.

II. Offline Procedures and Recovery Contract

The procedure descriptions here fix all behavior needed by the proofs. Rule ordering and witness choices may vary without changing the results.

A. Semantic compression

Given finite G, Σ and integers $T, N_b \geq 1$, start with $G' = G$. In Phase 1, process each rule γ whose head predicate is absent from its own body once, in any fixed order. On its turn, delete exactly $G_{\text{del}} = G' \cap \tau_{\{\gamma\},G'}$ from G'. The implementation may order strongly connected components and rule support counts, but the proof needs only this sequential deletion. In each of T Phase 2 rounds, partition the current G' into disjoint blocks $G^{(1)}_{\text{blk}}, \ldots, G^{(N_b)}_{\text{blk}}$ whose union is G'. Initialize $G_{\text{keep}} = G_{\text{cl}} = \emptyset$ and for $i = 1, \ldots, N_b$ update

$$G_{\text{keep}} \leftarrow G_{\text{keep}} \cup (G^{(i)}_{\text{blk}} \setminus G_{\text{cl}}), \qquad G_{\text{cl}} \leftarrow \Delta_{G_{\text{keep}},\Sigma}. \quad (2)$$

Replace G' by G_{keep} after each round and return the final graph as $G^\dagger$. There are no empty-body rules, so $\Delta_{\emptyset,\Sigma} = \emptyset$.

B. Record derivations

From $G^\dagger$, compute successive fixpoint levels $G^{(0)} = G^\dagger$ and $G^{(i)} = G^{(i-1)} \cup \tau_{\Sigma,G^{(i-1)}}$. At iteration i, let $G_{\text{known}} = G^{(i-1)}$ and let $G_{\text{new}} = G^\dagger$ for $i = 1$, or $G^{(i-1)} \setminus G^{(i-2)}$ otherwise. Enumerate rule instances whose body lies in G_{known} and whose body has at least one occurrence in G_{new}. For every newly derived head d, record one pair $\mathcal{D}[d] = (\gamma, \sigma)$ with $\sigma(H_\gamma) = d$ and $\sigma(B_\gamma) \subseteq G_{\text{known}}$. Ignore additional witnesses for the same head. The final $G^{(i)}$ is denoted $G^* = \Delta_{G^\dagger,\Sigma}$, and $\mathcal{D}$ is undefined on $G^\dagger$. The least iteration in which d appears is its derivation level. Let ℓ be the largest productive level.

Definition 1 (Complete one-hop transformed rule set)**.** Finite PD rules Σ^{cot} form a COT rule set for Σ relative to $G^\dagger$ if (i) $\Sigma \models \gamma$ for every $\gamma \in \Sigma^{\text{cot}}$, (ii) $\Delta_{G^\dagger,\Sigma} = G^\dagger \cup \tau_{\Sigma^{\text{cot}},G^\dagger}$, and (iii) Σ^{cot} contains an identity rule for each predicate in $G^\dagger$ or in a head of Σ^{cot}.

C. Generate one-hop rules

Initialize Σ^{cot} with identity rules for predicates occurring in G^* or Σ. Process each $d \in G^* \setminus G^\dagger$. If an existing rule instance over $G^\dagger$ already has head d, continue. Otherwise start from a freshly renamed copy of $\mathcal{D}[d]$ and its ground witness. While some indexed body occurrence has a witnessed fact outside $G^\dagger$, take that fact's recorded derivation, rename it apart, unify its head with the selected occurrence, and replace only that occurrence by its body. Apply the MGU to the entire resulting rule and factor the combined ground witnesses through the MGU. After every witnessed body fact lies in $G^\dagger$, identify duplicate body atoms and add the resulting rule. Finally deduplicate rules and repeatedly remove a non-identity rule subsumed by a remaining rule. This procedure is GenerateCOT, and its unfolding subroutine is UnfoldRule.

III. Query Evaluation Definitions

A conjunctive query (CQ) is $Q(\vec{X}) \leftarrow \alpha_1 \wedge \cdots \wedge \alpha_m$, where $m \geq 1$, $\vec{X}$ is an ordered tuple of answer variables in the body, and other variables are existential. It is Boolean when $\vec{X}$ is empty. A finite union of CQs (UCQ) has disjuncts of the same answer arity. Answers are ground matches, projected

onto $\vec{X}$. For atom α_k, let $\vec{Z}_k$ list its distinct variables. A candidate branch is obtained by freshly renaming a rule $\gamma \in \Sigma^{\mathrm{cot}}$, unifying H_γ with α_k, and using the unified body as a CQ with output mapping on $\vec{Z}_k$. The mapping enforces constants and identified variables. Let R_k^* be all matches of α_k over G^*.

The frontier ξ_k contains variables of α_k that occur in another query atom or in $\vec{X}$. The union of candidate branch outputs, projected to the ordered frontier, is R_k. Projection π, natural join $\bowtie$, and semi-join $\ltimes$ have ordinary set-relational meanings. Joins on no shared columns are Cartesian products. A zero-column relation is either $\{()\}$ or $\emptyset$. The base factorized plan P00 computes

$$\pi_{\vec{X}}(R_1 \bowtie \cdots \bowtie R_m). \tag{3}$$

Candidate-free groups have an empty relation. An empty projection is an existence test.

QFSE (query-frontier-sensitive execution) is the P10 switch. Within each candidate, order the body atom occurrences as $\beta_1, \ldots, \beta_s$. At stage j, retain the variables among the first j atoms that are required by the branch's mapped output or by a later body atom. Call this set $\xi_{k,i}^{(j)}$, where i identifies the candidate. Starting with $R_{k,i}^{(0)} = \{()\}$, join each atom's match relation R_{β_j} and project to this set. A semi-join may replace a join if no new required column is introduced. Required output bindings remain fixed across all stages.

UFA (unary frontier absorption) is the P01 switch. For a variable x with singleton-frontier source groups, set

$$\mathcal{I}_x = \{k : \xi_k = \{x\}\}, \qquad R_x^{\mathrm{u}} = \bigcap_{k \in \mathcal{I}_x} \pi_x(R_k). \tag{4}$$

Push this unary membership test into every group with a frontier of size greater than one that contains x, then remove the source groups from the top-level join. If no such group exists, retain the unary relation when x is an answer variable and retain its nonemptiness as an existence guard otherwise. Zero-frontier groups always remain existence guards. P11 applies both switches, placing unary tests on each candidate's mapped output before projection. COMPILE emits the corresponding unions, joins, semi-joins, projections, and guards as distinct, null-free SQL over $G^\dagger$. An offline policy tree chooses one of P00, P01, P10, and P11 for execution.

IV. COMPLETE PROOFS

Lemma 1 (Redundancy removal preserves the least fixpoint)**.** For finite G, Σ and $T, N_b \geq 1$, semantic compression returns $G^\dagger \subseteq G$ with $\Delta_{G^\dagger,\Sigma} = \Delta_{G,\Sigma}$.

Proof of Lemma 1. Consider one Phase 1 deletion step. Let $G' \subseteq G$ be the current graph and γ the non-recursive rule used in this step. The subgraph deleted in this step is $G_{\mathrm{del}} = G' \cap \tau_{\{\gamma\},G'}$. Every fact in G_{del} has the head predicate of γ, which occurs in no body atom of γ. Consequently, for each $d \in G_{\mathrm{del}}$, any body match in G' witnessing d remains a body match in $G' \setminus G_{\mathrm{del}}$. Thus $G_{\mathrm{del}} \subseteq \Delta_{G' \setminus G_{\mathrm{del}},\Sigma}$ and $G' \subseteq \Delta_{G' \setminus G_{\mathrm{del}},\Sigma}$. Since $G' \setminus G_{\mathrm{del}} \subseteq G'$, monotonicity and idempotence of least-fixpoint closure imply

$$\Delta_{G' \setminus G_{\mathrm{del}},\Sigma} = \Delta_{G',\Sigma}.$$

Applying this argument to every Phase 1 deletion proves that its output G' is a subset of G and has closure $\Delta_{G,\Sigma}$. The argument does not depend on the particular SCC or rule order.

For a Phase 2 round, let G' be its input graph and let $G'_i = \bigcup_{j=1}^{i} G_{\mathrm{blk}}^{(j)}$ be the subgraph formed by its first i blocks, with $G'_0 = \emptyset$. Denote the retained set and its closure after block i by $G_{\mathrm{keep}}^{(i)}$ and $G_{\mathrm{cl}}^{(i)} = \Delta_{G_{\mathrm{keep}}^{(i)},\Sigma}$, with $G_{\mathrm{keep}}^{(0)} = G_{\mathrm{cl}}^{(0)} = \emptyset$. The initial closure is empty because PD rules have nonempty bodies. We claim, by induction on i, that

$$G_{\mathrm{keep}}^{(i)} \subseteq G'_i \subseteq G_{\mathrm{cl}}^{(i)}.$$

It holds for $i = 0$. At block i, the update $G_{\mathrm{keep}}^{(i)} = G_{\mathrm{keep}}^{(i-1)} \cup (G_{\mathrm{blk}}^{(i)} \setminus G_{\mathrm{cl}}^{(i-1)})$ retains only input facts, so $G_{\mathrm{keep}}^{(i)} \subseteq G'_i$. Every fact of G'_{i-1} lies in $G_{\mathrm{cl}}^{(i-1)} \subseteq G_{\mathrm{cl}}^{(i)}$. A fact of $G_{\mathrm{blk}}^{(i)}$ either already lies in $G_{\mathrm{cl}}^{(i-1)}$ or is added to $G_{\mathrm{keep}}^{(i)} \subseteq G_{\mathrm{cl}}^{(i)}$. Hence $G'_i \subseteq G_{\mathrm{cl}}^{(i)}$, proving the induction step. At the last block, the partition gives $G'_{N_b} = G'$, so $G_{\mathrm{keep}}^{(N_b)} \subseteq G' \subseteq \Delta_{G_{\mathrm{keep}}^{(N_b)},\Sigma}$. Monotonicity and idempotence yield $\Delta_{G',\Sigma} = \Delta_{G_{\mathrm{keep}}^{(N_b)},\Sigma}$. Each round therefore preserves its input closure while retaining a subset of its input. Induction over the T rounds, following Phase 1, proves $G^\dagger \subseteq G$ and $\Delta_{G^\dagger,\Sigma} = \Delta_{G,\Sigma}$. All loops terminate because the graphs and rule set are finite and the function-free closure reaches a finite fixpoint. □

Theorem 1 (Correctness of one-hop rule construction)**.** Given finite G, Σ, let semantic compression return $G^\dagger$ and derivation recording return $G^*, \mathcal{D}$. The rule-generation procedure constructs a finite COT rule set Σ^{cot} independently of any subsequently submitted query.

Proof. We first justify the derivation binding returned by Section II-B. Define $G^{(0)} = G^\dagger$ and $G^{(i)} = G^{(i-1)} \cup \tau_{\Sigma,G^{(i-1)}}$ for $i \geq 1$. At the start of iteration i, the algorithm has $G_{\mathrm{known}} = G^{(i-1)}$ and, for $i > 1$, $G_{\mathrm{new}} = G^{(i-1)} \setminus G^{(i-2)}$; for $i = 1$, $G_{\mathrm{new}} = G^{(0)}$. Every head admitted by the algorithm belongs to $G^{(i)} \setminus G^{(i-1)}$, because all of its body facts belong to $G^{(i-1)}$ and previously known heads are excluded. Conversely, let $d \in G^{(i)} \setminus G^{(i-1)}$ and choose a rule instance deriving it from $G^{(i-1)}$. At least one body occurrence belongs to G_{new}. Otherwise all its body facts would already belong to $G^{(i-2)}$, and d would be in $G^{(i-1)}$. For $i = 1$, every body occurrence belongs to the initial G_{new}. The algorithm examines this position and therefore admits d. It records one witness for each newly admitted head, even if several instances derive the same head. Induction proves that the returned G^* is $\Delta_{G^\dagger,\Sigma}$ and that every $d \in G^* \setminus G^\dagger$ has a recorded witness whose body facts have strictly smaller derivation levels. Finiteness of the active domain makes this process terminate.

Now consider UNFOLDRULE for an omitted fact d. Maintain the following invariant for its current rule $\gamma^{(j)}$ and ground witness σ_{w}:

$$\Sigma \models \gamma^{(j)}, \qquad \sigma_{\mathrm{w}}(H_{\gamma^{(j)}}) = d, \qquad \sigma_{\mathrm{w}}(B_{\gamma^{(j)}}) \subseteq G^*.$$

It holds initially for the freshly renamed input rule recorded for d. If a body occurrence β has witness $e = \sigma_{\mathrm{w}}(\beta) \notin G^\dagger$, the

binding $\mathcal{D}[e]$ provides a freshly renamed rule γ' and ground witness σ' with $\sigma'(H_{\gamma'}) = e$. The two clauses have disjoint variables. Their ground witnesses agree on the resolved atom, so their union unifies β and $H_{\gamma'}$. The relevant MGU μ therefore factors this ground unifier on the resolved atoms. Extending the factorization by the original ground assignments on other clause variables gives the ground substitution λ used in the algorithm. Applying μ to the whole rule and replacing just the selected body occurrence by $\mu(B_{\gamma'})$ preserves the witnessed head and assigns the child's recorded body facts to the inserted occurrences. Definite-clause resolution also preserves $\Sigma \models \gamma^{(j+1)}$. Range restriction is preserved: a head variable whose only body occurrence was replaced remains represented in the range-restricted child's body. Thus the invariant continues to hold.

Each expanded occurrence has a derivation level greater than the levels of its inserted supports. Expansion therefore forms a finitely branching tree of depth at most ℓ, the number of productive derivation levels. It terminates with every ground body fact in $G^\dagger$. Keeping occurrences distinct until the end ensures that only the chosen occurrence is expanded. Removing duplicate body atoms on return changes neither the clause's meaning nor its ground witness. Hence the returned path rule γ_d is entailed by Σ and has a ground instance with head d and body contained in $G^\dagger$.

Every initialized identity rule is a tautology and covers its retained facts in one step; pruning retains these identity rules. During the loop in Section II-C, each omitted fact is either already covered by a current rule or receives such a path rule. Therefore, before pruning, every fact in G^* is covered by an instance over $G^\dagger$. Deduplication preserves this property. Suppose a remaining rule γ' subsumes a rule γ being removed, witnessed by a substitution θ with $\theta(H_{\gamma'}) = H_\gamma$ and $\theta(B_{\gamma'}) \subseteq B_\gamma$. For any ground witness σ of γ over $G^\dagger$, $\sigma \circ \theta$ witnesses the same head through γ', since

$$(\sigma \circ \theta)(B_{\gamma'}) \subseteq \sigma(B_\gamma) \subseteq G^\dagger.$$

Each pruning step therefore preserves coverage. All surviving rules are entailed by Σ, giving $G^\dagger \cup \tau_{\Sigma^{\mathrm{cot}},G^\dagger} \subseteq \Delta_{G^\dagger,\Sigma}$; coverage gives the reverse inclusion. Thus Σ^{cot} satisfies all conditions of Definition 1. At most one finite path rule is generated per omitted closure fact, in addition to finitely many identities, so Σ^{cot} is finite. None of these operations uses the query. □

Theorem 2 (Exact single-step query answering)**.** Under the preceding assumptions, for every CQ Q, each of P00, P01, P10, and P11 evaluated over $G^\dagger$ returns exactly the answers to Q over G^*. Policy-tree selection preserves this equality. If G is rule-closed, these are the answers over the original G. The statement also holds for finite UCQs by disjunct-wise union.

Proof. Lemma 1 gives $G^* = \Delta_{G^\dagger,\Sigma}$, and Theorem 1 gives $G^* = G^\dagger \cup \tau_{\Sigma^{\mathrm{cot}},G^\dagger}$. The initialized identities cover every fact of $G^\dagger$, and the pruning argument in the preceding proof preserves that coverage. Consequently, the stronger atomic equality

$$\tau_{\Sigma^{\mathrm{cot}},G^\dagger} = G^*$$

holds. This equality is useful because atom-wise rewriting uses rule candidates, rather than a separate stored-fact branch.

Fix a query atom α_k with distinct variables $\vec{Z}_k$. Let R_k^* be its match relation over G^*, with columns $\vec{Z}_k$. For any candidate rule, a body match over $G^\dagger$ produces an instance of its head in G^* by soundness. Unifying that head with α_k therefore produces a tuple in R_k^*; the output mapping preserves constants and repeated variables. Conversely, take any tuple in R_k^* and its ground match of α_k. By the atomic equality, a ground instance of some $\gamma \in \Sigma^{\mathrm{cot}}$ has the same head fact and a body match in $G^\dagger$. Rename γ apart from the query. The two head matches give a ground unifier of H_γ and α_k. Relevant-MGU factorization yields a match of the corresponding candidate body, extending the ground substitution to body-only rule variables using the original rule-instance witness. Its output tuple is precisely the selected tuple of R_k^*. Hence the union of candidate branches equals R_k^*.

The query frontier ξ_k contains every variable of α_k that either occurs in another query atom or is an answer variable. All other variables of α_k can be existentially quantified within this group. Therefore its compiled relation is $R_k = \pi_{\xi_k}(R_k^*)$. By ordinary CQ semantics, the answer relation for Q over G^* is

$$\pi_{\vec{X}}\big(R_1 \bowtie \cdots \bowtie R_m\big).$$

This also holds for disconnected atoms, constants, and identified variables. With an empty frontier, R_k is either the unit zero-column tuple or the empty relation, so it contributes exactly the required existence condition. The same convention handles Boolean queries. This proves P00.

For QFSE, fix any ordering $\beta_1, \ldots, \beta_s$ of a candidate body's atom occurrences. The retained variable set $\xi_{k,i}^{(j)}$ in Section III contains exactly the variables among the first j atoms needed by the output or a later atom. The stage invariant is

$$R_{k,i}^{(j)} = \pi_{\xi_{k,i}^{(j)}}(R_{\beta_1} \bowtie \cdots \bowtie R_{\beta_j}),$$

with $R_{k,i}^{(0)} = \{()\}$. A variable projected out at stage $j-1$ occurs neither in the branch's required output nor in any remaining atom; hence it cannot influence a later join. Joining $R_{k,i}^{(j-1)}$ with R_{β_j} and projecting onto $\xi_{k,i}^{(j)}$ establishes the invariant by induction. If every variable in $\xi_{k,i}^{(j)}$ is already a column of $R_{k,i}^{(j-1)}$, the current atom introduces no required output variable. Under set semantics,

$$\pi_{\xi_{k,i}^{(j)}}(R_{k,i}^{(j-1)} \bowtie R_{\beta_j}) = \pi_{\xi_{k,i}^{(j)}}(R_{k,i}^{(j-1)} \ltimes R_{\beta_j}),$$

so the permitted semi-join implements the same stage. Constant and repeated-variable constraints belong to each atom's match relation. At the final stage, the retained variables and any constant output terms restore exactly the original branch's ξ_k columns. Thus QFSE preserves every group, regardless of the chosen body-atom order.

For UFA, each source group with $\xi_k = \{x\}$ is a unary relation on x. Their natural join is exactly $R_x^{\mathrm{u}} = \bigcap_{k \in \mathcal{I}_x} \pi_x(R_k)$. For a multi-frontier group R_j containing x, replacing it by

$R_j \ltimes_x R_x^{\mathrm{u}}$ pushes the same membership constraint into that group. If at least one such group remains, its x column keeps the constraint in the top-level join, so the original singleton-frontier sources can be removed. Applying the filter to every eligible group is idempotent under set semantics. If no multi-frontier group contains x, retaining R_x^{u} when x is an answer variable preserves its possible output values; when x is existential, an $\mathrm{EXISTS}(R_x^{\mathrm{u}})$ guard preserves its required nonemptiness. Every zero-frontier group likewise remains an existence guard. These rules also cover an empty top-level join.

In P11, the R_x^{u} membership test is applied to each candidate using its mapped output value, after that value is bound or immediately if it is a constant. A frontier variable needed by the test is included in the fixed required output of QFSE and cannot be projected away before or after the test. Thus the QFSE and UFA equivalences compose. The compiler of Section III emits the unions, joins, projections, semi-joins, and existence guards just proved equivalent, with distinct outputs under the set semantics defined here. Its SQL interpretation uses a faithful null-free encoding of $G^\dagger$ and equality joins. For completed executions, each of P00, P01, P10, and P11 therefore returns exactly the answers to Q over G^*. Selecting one of these equivalent modes with the cost-sensitive policy tree cannot change the answer set. Taking a finite union of CQs with a common answer schema also preserves the equality branch by branch. □

V. Auxiliary Size and Cost Bounds

Let $M = |G^* \setminus G^\dagger|$, let P count the initialized identity rules, and put $b = \max(\{1\} \cup \{|B_\gamma| : \gamma \in \Sigma\})$. Let $L_{\Sigma^{\mathrm{cot}}}$ count head and body atoms in the final rule set. The size bound

$$L_{\Sigma^{\mathrm{cot}}} \le 2P + M(1 + b^\ell) \tag{5}$$

follows from decreasing derivation levels. Each productive iteration of Section II-B creates at least one omitted fact, so $\ell \le M$. A path rule's unfolding tree has branching at most b and depth at most ℓ. It has at most b^ℓ leaf occurrences, hence at most that many body atoms after deduplication. There are at most M generated path rules, each with one head, and P identities, each with one head and one body atom. Deduplication and subsumption removal can only decrease this count. The estimate establishes finiteness; it is exponential in ℓ when $b > 1$.

The grouped candidate representation stores one group record per query atom, one record per candidate, and each candidate body once. With unary/binary predicates and interned symbols, the output mapping has constant size per candidate. For m atoms, n_k candidates in group k, and total candidate-body length L, this gives $O(m + \sum_k n_k + L)$ words. The bound concerns the grouped representation, not emitted SQL or database intermediate results. Compiling four modes, extracting their features, traversing a height-h policy tree, and executing one selected plan gives

$$\begin{aligned} C_{\mathrm{online}}(Q) = O\big(&4C_{\mathrm{comp}}(Q) + C_{\mathrm{feat}}(Q) \\ &+ h + C_{\mathrm{query}}(Q)\big), \end{aligned} \tag{6}$$

where the three C terms bound one-mode compilation, feature extraction, and selected-plan execution, respectively. Counting the retained facts, the head and body atoms of Σ^{cot}, and the policy model gives

$$S_{\mathrm{stored}} = O(|G^\dagger| + L_{\Sigma^{\mathrm{cot}}} + S_{\mathrm{policy}}) \tag{7}$$

words under bounded arity and interned symbols; S_{policy} is the policy-tree size. Neither storage nor representation bounds include execution intermediates.